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MILITARY SPECIFICATION

BLAST CLEANING MACHINES

This specification is approved for use by HQ AFLC CASO/LODS, Federal Center Battle Creek, MI 49016, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

*1.1 Scope. This specification covers dry blast machines for surface cleaning, surface conditioning and peening of ferrous and nonferrous metals using a blasting media as specified in 3.4.4.

*1.2 Classification. The type machines covered by this specification shall be of the following types and sizes as specified (see 6.2).

Type I - Cabinet, Stationary
Size 1
Size 2
Size 3

Type II - External hand-held gun

2. APPLICABLE DOCUMENTS.

*2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

J-C-175	Cable Assembly, Power, Electrical, For 125 Volt, 250 Volt, and 125/250 Volt, 50-60 Hz, Equipment
W-C-596	Connector, Electrical, Power, General Specification For
PPP-B-636	Box, Shipping, Fiberboard
PPP-T-60	Tape, Packaging, Waterproof

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: AFLC CASO/LODS, Federal Center, Battle Creek, MI 49016 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4940

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Military

MIL-P-116	Preservation-packaging, Methods of
MIL-S-851	Steel Grit, Shot, and Cut Wire Shot, and Iron Grit and Shot - Blast Cleaning and Peening
DOD-D-1000	Drawings, Engineering and Associated Lists
MIL-G-9954	Glass Beads, For Cleaning and Peening
MIL-C-17504	Coating Compound, Acrylic, Clear
MIL-A-21380	Abrasive Material, For Blasting
MIL-C-52950	Crate, Wood, Open and Covered

STANDARDS

Federal

FED-STD-H28	Screw Thread Standard for Federal Services
FED-STD-595	Colors

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-808	Finish, Protective, and Codes , For Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute Inc

* S1.4 General Purpose Sound Level Meters

(Application for copies should be addressed to the American National Standards Institute, Inc; 1430 Broadway, New York, NY 10018.)

NATIONAL ELECTRICAL MANUFACTURER'S ASSOCIATION

NEMA Standards

(Application for copies should be addressed to the National Electrical Manufacturer's Association, 155 East 44th Street, New York, NY 10017.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Rules For the Construction of Unfired Pressure Vessels

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(Copies of ASME publications may be obtained from the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

3. REQUIREMENTS.

3.1 Preproduction testing. This specification makes provision for preproduction testing.

3.2 Components.

*3.2.1 Type I. The Type I machine shall consist of the following major components:

Item No.	Item or System Name
1	Cabinet
2	Exhaust and Dust Collector System
3	Reclaim System
4	Abrasive Hopper
5	Blast Gun and Hose Assembly
6	Air and Media Operating Controls
7	Air Pressure Regulator and Gage
8	Moisture Separator
9	Instructional Panel

*3.2.2 Type II. The Type II machine shall consist of the following major components:

Item No.	Item or System Name
1	Recovery and Dust Collector System
2	Reclaim System
3	Abrasive Hopper
4	Blast Gun and Hose Assembly
5	Casters
6	Air Pressure Regulator and Gage
7	Moisture Separator
8	Instructional Panel
9	Required Accessories

3.3 Preproduction sample. The contractor shall furnish a preproduction sample for inspection and tests to determine conformance with this specification. Approval of the preproduction sample, by the activity concerned, shall not relieve the contractor of his/her obligation to supply equipment conforming to this specification. The preproduction sample shall be identical with the proposed production items and shall be made and assembled by the same methods that are to be used in production of the items to fulfill the terms of the contract. No change or deviation from the preproduction sample will be acceptable without prior written approval from the contracting officer.

3.4 Materials. Materials shall be as specified herein. Materials not specifically designated shall be of standard commercial quality.

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3.4.1 Standard parts. MIL Standard parts shall be used where applicable. In the event MIL Standard parts are not available, commercial parts such as screws, nuts, bolts, et cetera, may be used provided they conform to the requirements of this specification.

*3.4.2 Protective treatment. Materials that are subject to deterioration upon exposure to climatic and environmental conditions during use, shall be given a protective treatment which shall not hinder the performance of the machines as required by this specification. The use of any protective coating that shall crack, chip, or scale with age or extremes of climatic environmental conditions shall be avoided.

3.4.3 Dissimilar metals. Unless suitably protected, dissimilar metals as defined in MIL-STD-889 shall not be used in intimate contact with each other.

*3.4.4 Blasting media. The blast machines shall be capable of satisfactory performance using any blasting media of the following materials: Glass Beads conforming to MIL-G-9954, Nominal G110 to G230 Steel Shot and Steel Grit conforming to MIL-S-851 and Aluminum Oxide, Silicone Carbide, Garnet, and Emery conforming to MIL-A-21380.

*3.5 Design. The machines shall be designed for simplicity of operation and safety to operating personnel and shall permit easy accessibility of components for normal maintenance and servicing. A suction or pressure type feed shall supply blasting media to the blast gun where high pressure air under controlled conditions propels the blasting media at high velocity in a uniform and consistent stream to the parts being cleaned or conditioned. The machine shall be capable of blasting continuously for periods in excess of 7 minutes. The blast shut-off time required for automatic recharging of the abrasive hopper shall not exceed 30 seconds. The machine shall be designed to accommodate blasting media as specified in 3.4.4.

*3.5.1 Type I. The Type I blast machine shall be so designed that the operator directs the blast media upon the workpiece while standing outside the cabinet and extending his arms into the cabinet through protective openings. The Type I machine design shall include an exhaust and dust collector system for maintaining a dust free climate within the cabinet. A reclaim system shall remove reuseable blasting media from the dust stream and return the blasting media to the abrasive hopper. The dust shall be collected in a cloth filter.

*3.5.2 Type II. The Type II blast machine shall be designed for cleaning objects that are not suitable for cleaning within a cabinet type machine. The blast gun shall be so designed that the operator can clean flat, round, cylinderical, and square objects without loss of blasting media to the atmosphere. The Type II machine design shall include a recovery system to return spent blasting media from the workpiece to the unit and a reclaim system to separate reuseable blasting media from the air stream and return it to the abrasive hopper. The useable blasting media and dust shall be collected in a cloth filter. The blast generating unit shall be mounted on casters to afford easy maneuverability.

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*3.6 Construction. The dry blast machine shall be complete so that when installed it can be used for the operations specified herein. All wiring and electrical controls shall be installed with all connections completed within and on the machine, so that upon installation of the machine, all that is required is connection to a pneumatic supply and a power source. The machine shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.6.1 Castings and forgings. All castings shall be free from blowholes; porosity, hardspots, shrinkage defects, cracks, and other defects. Forgings shall be free from scale, inclusions, cold shuts, mismatching and other defects that may affect the structural strength of the forgings involved. Strength and other essential physical and chemical properties of the castings and forgings shall be adequate to meet the performance requirements specified herein.

*3.6.2 Welding, brazing and soldering. Welding, brazing, and soldering shall be performed in accordance with good commercial practice. In no event shall such processes including peening or plugging be used on castings or forgings for reclaiming any parts of the machine.

3.6.3 Fastening devices. All screws, pins, bolts and similar parts shall be installed with means for preventing loss of tightness and adjustment. All such parts subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.6.4 Electrical connections. Electrical connections shall be mechanically secured and soldered or brazed, except connections on screw type terminals which shall be mechanically secured. Resin base fluxes shall be used in all soldering.

*3.6.5 Electrical components. Electrical components of the dry blast machines including starters, relays, switches, wiring and their enclosures shall conform to and shall be located in accordance with the NEMA Standards except as specified herein. No component part shall be used in any application where the requirements exceed the manufacturer's rating for the part.

*3.6.6 Threads. All threaded parts shall be in the English System and shall comply with FED-STD-H28.

*3.6.7 Instruments and controls. All instruments and gages supplied on the dry blast machines shall correspond in appearance. All instruments, switches, controls, and outlets shall be plainly identified by permanent plates.

3.6.8 Lubrication. All bearings (except sealed-for-life-type) and other working parts shall be provided with means for lubrication.

3.6.9 Pressure vessels. Pressure vessels shall be designed and constructed in accordance with the ASME Rules For Construction of Unfired Pressure Vessels.

3.7 Performance.

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*3.7.1 Type I. The Type I dry blast machine shall be capable of regulating a pneumatic supply pressure of 125 pounds per square inch gage to an operating pressure of 20 to 100 pounds per square inch gage. The machine shall be capable of peening to the intensities specified in Table I with the designated air pressure, bead size, and peening time. The blasting media shall be reclaimed, cleaned, and recycled. When blasting at 80 pounds per square inch gage with size 13 glass beads, the machine shall produce a cleaned path not less than 1 inch wide at a rate of 1 lineal foot in 10 seconds. When blasting at 80 pounds per square inch gage with 26 grit aluminum oxide, the machine shall produce a cleaned path not less than 1 inch wide at a rate of not less than 4 lineal feet in 10 seconds. The noise level at the operators position shall not exceed 84 decibel A slow scale.

*3.7.1.1 Media rate or flow. The dry blast machine shall be capable of circulating a minimum of 100 pounds of blasting media per hour at the discharge nozzle without evidence of strain or damage to any component of the machine.

TABLE I

PERFORMANCE CHARACTERISTICS

	BEAD SIZE	AIR PRESSURE AT REGULATOR (POUNDS PER SQUARE INCH GAGE)	TIME	PEENING INTENSITY (MINIMUM) ALMEN ARC HEIGHT
	3	40	22 sec	.006 ± .0005 "A"
*	3	60	22 sec	.010 ± .0005 "A"
	3	80	22 sec	.014 ± .0005 "A"
	8	40	22 sec	.0025 ± .0005 "A"
	8	60	22 sec	.004 ± .0005 "A"
	8	80	22 sec	.0045 ± .0005 "A"
	12	40	22 sec	.0015 ± .0005 "A"
	12	60	22 sec	.002 ± .0005 "A"
	12	80	22 sec	.0025 ± .0005 "A"

*3.7.1.2 Blasting media capacity. The dry blast machine shall have a minimum blasting media capacity to maintain the rate of flow specified in 3.7.1.1. The machine shall be designed so that it can be easily loaded and unloaded of blasting media by one man.

*3.7.2 Type II. The Type II blast machine shall be capable of regulating a pneumatic supply pressure of 125 pounds per square inch gage to an operating

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pressure of 20 to 100 pounds per square inch gage. It shall be capable of discharging a constant, evenly distributed stream of blasting media on a material surface having a minimum curvature of 12 inch radius and shall reclaim, clean and recycle the blasting media during cleaning operations. When blasting at 80 pounds per square inch gage with size 13 glass beads, the machine shall produce a cleaned path not less than 1 inch wide at a rate of 1 lineal foot in 10 seconds. When blasting at 80 pounds per square inch gage with 26 grit aluminum oxide, the machine shall produce a cleaned path not less than 1 inch wide at a rate of 4 lineal feet in 10 seconds. In addition to normal cleaning operations, the machine shall be capable of peening to the intensities specified in Table I at the designated conditions. The blast gun shall operate at a minimum distance of 20 feet from the caster mounted blast generator.

*3.7.2.1 Media rate of flow. The external hand-held dry machine shall be capable of circulating a minimum of 100 pounds of blasting media per hour at the discharge nozzle without evidence of strain or damage to the external hand-held dry machine and its components.

*3.7.2.2 Blasting media capacity. The external hand-held dry machine shall have a minimum blasting media capacity to maintain the rate of flow specified in 3.7.2.1. The external hand-held dry machine shall be designed so that it can be easily loaded and unloaded of blasting media by one man.

3.8 Detail of components.

3.8.1 Type I.

3.8.1.1 Cabinet. The cabinet shall be of welded construction; the panels shall be constructed from sheet metal not less than .1046 inch nominal thickness (Manufacturer's Standard Gage Number 12) and shall be dust tight. The cabinet frame and supporting members shall be located, proportioned and braced so as to support the cabinet and all its components. The cabinet size shall be as follows:

Size 1 - Work area 4 feet wide by 3 feet deep by 3 feet high
Size 2 - Work area 5 feet wide by 4 feet deep by 3 feet high
Size 3 - Work area 6 feet wide by 6 feet deep by 5 feet high

3.8.1.1.1 Access door. The access door for insertion of work piece shall provide the maximum area possible without hindering or impairing the structure design of the cabinet. The door of the size 1 cabinet shall be hinged type construction with a positive type latch. The door shall be of rigid construction with a seal to prevent escape of dust or blast media. The size 2 and size 3 cabinet door shall be of the vertical lift type having an opening not less than 52 inches by 52 inches. It shall employ a dead weight counterbalance system, and it must operate in a tract guide with antifriction bearings. The door shall be of rigid construction with a seal to prevent the escape of dust or blast media.

3.8.1.1.2 Work floor. The work floor of the cabinet shall be constructed of 3/16 inch perforated steel plate coated to a thickness of not less than 15 mils with a resilient material. The floor shall be capable of supporting a

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500 pound load with a maximum allowable deflection of 1/4 inch. The floor shall be supported within the cabinet in such a manner that it can be easily removed and replaced.

*3.8.1.1.3 Observation window. The observation window shall be located above the arm ports and at a height as to permit the operator full view of the area covered by the abrasive discharge. The window shall be constructed of safety glass (either plastic laminated or tempered). The inside surface of the window shall be protected from abrasive erosion by screening or transparent plastic film. The window shall have a minimum dimension of 23 inches long by 8 inches high. The glass shall be perpendicular to operator's line of sight.

*3.8.1.1.4 Lighting system. The work area of the cabinet shall be illuminated by an external fluorescent or incandescent light. The lgiht shall provide a minimum allowable light intensity of 200 foot candles at the center of the cabinet at the work level and 150 foot candles at the corners of the cabinet at the work level. The light shall be shielded from the operator's vision.

3.8.1.1.5 Working ports. Working ports or arm holes shall be provided for access to the blasting chamber for the purpose of operating the blasting gun and perfroming the cleaning operation. The ports shall be positioned in the front of the cabinet and shall be of such size and location to enable the operator to perform sustained cleaning operations without undue fatigue. The operator shall be provided arm and hand protection that will be completely sealed to prevent escape of blast media and dust. The arm and hand protection shall permit a minimum 24 inch reach to operator and shall in no way hinder the movement of arms and hands during the blasting operation. The arm and hand protection shall be easily removed and replaced.

*3.8.1.1.6 Cabinet accessories. The accessories described in 3.8.1.1.7 and 3.8.1.1.8 shall be furnished with Type I, Sizes 2 and 3 machines.

3.8.1.1.7 Loading car track. Track and extension assemblies shall support the loading car described in 3.8.1.1.8 plus a minimum workload of 150 pounds per square foot of car loading area. Track shall be both inside and outside the blast cabinet. The inbound end of external track shall be supported by bolting to the cabinet. The length of the external track shall be at least that of the internal track.

3.8.1.1.8 Loading car and turntable. The loading car shall have a cast iron or fabricated steel frame, and four grooved or flanged type wheels. The car shall roll on the tracks described in 3.8.1.1.7. A bearing mounted turntable of at least 24 inches diameter for the size 2 cabinet and at least 48 inches diameter for the size 3 cabinet shall be installed in the center of the loading car. The working surface of the turntable shall be coated with a thickness of not less than 15 mils of resilient material. The car and turntable shall safely withstand a workload of 150 pounds per square foot of car loading area. The distance from the top of the turntable to the top of the car tracks shall not exceed 7 inches. Car wheel and turntable bearings

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shall be sealed against dust and abrasive contamination, and shall turn freely when under load, after continued use in the blast cabinet.

*3.8.1.2 Exhaust and dust collector system. The exhaust and dust collector system shall maintain the inside of the cabinet free of dust while the operator is blast cleaning. The exhaust system shall be of sufficient capacity to replace the air within the cabinet not less than 10 times per minute when discharging against a static discharge pressure equivalent to 6 inches of water. The dust shall be collected within a fire proof cotton sateen cloth filter that has ample capacity for both the air volume and dust load involved. If the bag or stocking type dust collector is used, the surface area shall not be less than 52 square feet. The filter bag shall be easy to remove, open, and clean. One additional spare bag or filter shall be provided with each machine.

*3.8.1.3 Reclaim system. The blast machine shall be equipped with a cyclone separator type reclaim system which shall separate reuseable abrasive from exhaust air stream and return it to the abrasive hopper. The reclaim system shall not be less than 80 percent effective.

3.8.1.4 Abrasive hopper. The abrasive hopper shall have a capacity of not less than 40 pounds of size 3 glass beads and shall have provisions for draining the charge of beads. The abrasive hopper shall continuously receive reuseable abrasive as discharged by the reclaim system. All parts of the hopper shall be readily accessible for repair.

3.8.1.5 Blast gun and hose assembly. The blast gun shall be designed to provide optimum performance under this document. The nozzle and all internal passages of the gun assembly in contact with the abrasive stream shall be protected by a minumum of 1/8 inch thick tungsten carbide or boron carbide. The nozzle shall be readily replaceable. The hose shall be of standard commercial grade and of such length and flexibility that the operator can easily clean material throughout the range of the cabinet.

3.8.1.6 Air and media operating controls. The air and media operating controls (start and stop media flow) shall be regulated by foot treadle or knee contactor.

*3.8.1.7 Air pressure regulator and gage. A 150-pound maximum variable pressure regulator with gage shall be furnished to regulate a 125 pounds per square inch gage air supply to a working pressure of 20 to 100 pounds per square inch gage. The pressure regulator and gage shall be mounted in a readily accessible location.

*3.8.1.8 Moisture separator. A moisture separator shall be installed on the machine ahead of all air operated components. The moisture separator shall be capable of removing at least 95 percent of the condensate from the air supply. The separator shall include a screen to remove other contaminants from the air supply line which may be harmful to the air operated components. The moisture separator shall have a clean shatter resistant plastic globe with a drain valve.

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*3.8.1.9 Instructional panel. Brief and easily readable operating and precautionary instructions shall be permanently affixed to the blast machine in a readily accessible and protected location. The instructions shall be clear, concise, and sufficient to enable operation of the machine without damage to the equipment or injury to operating personnel. The instruction panel shall be made of sheet aluminum or sheet zinc of not less than 0.050 inch thickness, anodized or etched to produce raised markings with a black or other dark color background, and with a raised border of not less than 1/4 inch width.

3.8.1.10 Cleaning tools. An air hose with nozzle shall be furnished within the cabinet. This air hose with nozzles is intended for blowing blast media from work pieces upon completion of blast cleaning. A hangar shall be furnished within the cabinet for storage of the air hose and nozzle. A vacuum hose and suitable nozzle shall be furnished of length adequate to clean the floor four feet from the front and both sides of the machine. Means shall be furnished for convenient stowage of the vacuum hose and nozzle.

3.8.2 Type II.

*3.8.2.1 Recovery and dust collector system. The Type II blast machine shall be equipped with a recovery and dust collector system to prevent the escape of blasting media and dust to the atmosphere. The system shall pick-up spent abrasive at the workpiece and return it to the blast generator. The recovery system shall be not less than 90 percent efficient. The unuseable abrasive and dust shall be collected in the dust collector. The dust collector shall consist of cloth filter elements within a rectangular enclosure fabricated of either aluminum alloy or steel sheet. The filter elements shall be cloth multiple envelope type, each envelope being independently replaceable but shall be mounted as a set in a rigid frame, such that the complete set may be lifted free of the enclosure for periodic maintenance. A manually operated shaker mechanism shall be provided to permit shaking the filter to remove dust. The removed dust shall be collected on a tray at the bottom of the filter enclosure. Means shall be provided for easy removal of the dust tray. The recovery system vacuum pump shall be electrically driven and shall operate independently of the blast control. The recovery system on/off switch shall be mounted on the blast generator.

*3.8.2.2 Reclaim system. The Type II blast machine shall be equipped with a cyclone type reclaim system which shall separate reuseable abrasive from the spent abrasive returned from the workpiece by the recovery system. The reclaimed abrasive shall be returned to the abrasive hopper. The reclaim system shall be not less than 80 percent effective.

3.8.2.3 Abrasive hopper. The abrasive hopper shall have a capacity of not less than 5 pounds of size 3 glass beads and shall have provisions for draining the charge of beads. The hopper shall continuously receive reuseable abrasive as discharged by the reclaim system. All parts of the hopper shall be readily accessible for repair.

3.8.2.4 Blast gun and hose assembly. The blast gun shall be a hand-held unit designed and constructed to provide easy maneuverability by one operator

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without causing undue fatigue. The blast gun nozzle shall be readily replaceable and shall be made of tungsten carbide or boron carbide. The nozzle end of the blast gun shall be constructed to allow easy snap-on of the brushes listed in 3.8.2.9. A control valve shall be mounted on the blast gun to control the blasting operation. The control valve shall have a palm-actuated lever not less than 3 inches in length. The valve shall be spring loaded in the normally closed position. The blast gun shall be supplied with 20 feet of hose. The vacuum return, abrasive, and air hoses shall be standard commercial grade designed and selected for this application.

*3.8.2.5 Casters. The blast machine shall be mounted on industrial type casters to afford easy maneuverability by one man. A two wheel configuration is acceptable. If a four wheel design is used, locking type casters, two of which must swivel, and a towbar shall be required.

3.8.2.6 Air pressure regulator and gage. Same as 3.8.1.7.

3.8.2.7 Moisture separator. Same as 3.8.1.8.

3.8.2.8 Instructional panel. Same as 3,8.1.9.

*3.8.2.9 Required accessories. All accessories necessary for efficient operation of the machine, even though not specified herein, shall be supplied with the unit. The blast machine shall require only connection to shop air and electrical power. The following spare brushes shall be furnished with each machine: six outside and six inside corner brushes, six flat surface brushes, and six brush retainers.

3.8.2.9.1 Brushes. All brush shall be manufactured to definite standards, tolerances and clearances in order that they can be readily attached to the blast gun. The brushes shall be constructed of 0.014 inch diameter crimped nylon bristles, and shall be designed and manufactured in such a manner as to retain the abrasive without spillage to the atmosphere during blasting operations.

*3.8.2.9.2 Storage compartment. The Type II blast machine shall be furnished with a convenient storage compartment of adequate size to hold the spare brushes specified in 3.8.2.9. The storage compartment shall be constructed of either aluminum alloy or steel sheet and shall be permanently mounted on the machine. The storage compartment shall be weather-tight and shall have a hinged opening to permit easy access to the items contained herein.

*3.9 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of DOD-D-1000 shall govern the manufacturer's part numbers and changes hereto.

*3.10 Service conditions. The machine shall operate from a single electrical connection with a power cable conforming to Type I of J-C-175 no less than 8 feet long and equipped with a ground type plug conforming to the dimensions of W-C-596. Operating voltage shall be either 3 wire, single phase, 110 volts plus or minus 10 percent or 4 wire, three phase, 220 volts, plus or

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minus 10 percent with a stepdown transformer to supply single phase 110 volts for the light. The air supply shall be 125 pounds per square inch gage.

*3.11 Finish and painting. The machine shall be finished and painted in accordance with MIL-STD-808, F-100. The type I cabinet interior shall be white, color number 37886, and the exterior shall be gray, color number 16473 of FED-STD-595. The Type II machine shall be gray, color number 16473 of FED-STD-595.

*3.12 Identification of product. The machine shall be marked for identification in accordance with MIL-STD-130.

3.12.1 Nameplates and instruction plates. Nameplates and instruction plates shall be covered with one coat of clear plastic coating conforming to MIL-C-17504.

3.13 Workmanship. Workmanship shall be first class to assure procurement of good quality equipment of a neat general appearance. Standards of workmanship shall be such that equipment specified herein shall comply with the requirement in Section 3. All components, subassemblies, parts and final assembly shall comply with the requirements of Section 3. All components, subassemblies, parts and final assembly shall comply with tests in Section 4.

*3.14 High wear protection. The internal surfaces of the machine normally considered to be subjected to high wear from abrasive impact, due to flow direction changes, et cetera, shall be coated with an abrasive resistant material of the type normally protected by the manufacturer.

4. QUALITY ASSURANCE PROVISIONS.

*4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his/her own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

*4.2 Classification of tests. The inspection and testing of the blast machines shall be classified as follows:

a.	Preproduction test	see 4.3
b.	Acceptance test	see 4.4

4.3 Preproduction testing. The preproduction tests shall consist of all the tests described under 4.5 except the individual acceptance test specified in 4.5.7.

*4.3.1 Preproduction test sample. The preproduction test sample shall consist of one machine representative of the production equipment.

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*4.3.2 Preproduction test report. After the contractor completes the preproduction tests, he shall prepare a preproduction test report in accordance with MIL-STD-831 and furnish a copy to the procuring activity for approval. The report shall include 8 by 10 inch photographs showing each side of the machine. The report shall include the manufacturer's top (assembly) drawing of the machine. The test report shall include the completed examination of product and funtional check lists.

*4.4 Acceptance test. Each production item shall be subjected to an acceptance test consisting of the following tests as described under 4.5.

a.	Examination of product	4.5.1
b.	Mechanical inspection	4.5.2
c.	Functional check	4.5.3
d.	Pressure test	4.5.4
e.	Dielectric strength test	4.5.5
f.	Individual acceptance test	4.5.7
g.	Noise test (Type I)	4.5.8

*4.5 Test methods.

*4.5.1 Examination of product. The blast machine shall be examined to determine compliance with the requirements of this document. The examination shall be conducted using a manufacturer developed check list that shall list each specification characteristic, method of verification and results obtained.

*4.5.2 Mechanical inspection. A mechanical inspection of all components and parts of the blast machine for defects and damage shall be conducted. All pertinent data concerning condition, defects in workmanship, and damage through use prior to test shall be recorded.

*4.5.3 Functional check. All mechanical parts of the blast machine shall be checked to determine if each operates freely and functions properly. The functional check shall be conducted using a manufacturer developed list that shall show functions checked, how checked and results obtained.

*4.5.4 Pressure test. Each component and circuit of the blast machine that operates under positive pressure in normal service shall be hydrostatically (or pneumatically or with carbon dioxide gas at the contractor's option) pressure tested at 1-1/2 times working pressure for not less than 10 minutes. Distortion, damage, or leaks resulting from the pressure testing shall be cause for rejection of the components affected.

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*4.5.5 Dielectric strength test. The electrical system shall be subjected to a dielectric strength test at a potential equal to twice the working voltage plug 1,000 volts for a period of one minute. Any arcing, flash-over or charring of the insulation shall cause rejection of the machine.

*4.5.6 Performance test. The blast machine shall be subjected to a 4-hour performance test using blasting media conforming to the requirements of 3.4.4. During the test period, the machine shall be used to perform cleaning tasks that will demonstrate the cleaning capability of the unit. The contractor will provide the exhibits to be cleaned, which shall be irregular shapes and shall present cleaning problems typical to the application of this type equipment. While blasting at 80 pounds per square inch gage on a flat surface, the machine shall produce a cleaned path not less than one inch wide at the rate of one lineal foot in 10 seconds using size 13 glass beads and at the rate of 4 lineal feet in 10 seconds using 26 grit aluminum oxide. In addition to the cleaning tasks performed, Almen strips shall be peened to show compliance with Table I as specified.

*4.5.6.1 Reclaim system test. The reclaim system test shall be performed using Size 8 glass beads as the blasting media. The blasting media shall be sieved to insure a sieve analysis conforming to the limits set forth in MIL-G-9954. A minimum of 5 pounds of the blasting media shall be cycled through the blast machine at least 10 times while blasting at 80 pounds per square inch gage against a flat steel plate. The media will then be drained from the machine and weighed, the weight being recorded. After weighing, the media will again be sieved and the media retained on the sieve shall weigh not less than 80 percent of the weight recorded above. Values less than 80 percent shall cause rejection of the machine.

*4.5.6.2 Exhaust and dust collector test (Type I). During the 4-hour performance test of the Type I blast machine, the exhaust and dust collector system shall be observed for proper operation. Certification shall be considered as evidence that the exhaust system replaces the air within the cabinet at least 10 times per minute. However, a dust free and leak proof system shall still be required and proven to exist. The dust bag shall be removed, emptied, and replaced one time.

*4.5.6.3 Recovery and dust collector test (Type II). During the 4-hour performance test of the Type II blast machine, the recovery and dust collector system shall be observed for proper operation. The contractor shall demonstrate the recovery system to be 90 percent efficient. The blast gun brushes shall not allow escape of blast media to the atmosphere. A dust free climate shall exist around the blast machine. The mechanical filter shaker shall be operated and the dust tray removed and emptied.

4.5.6.4 Work floor test (Type I). The work floor shall be loaded at the center with 500 pounds and the maximum deflection measured. The deflection shall not exceed 1/4 inch.

4.5.6.5 Lighting system test (Type I). Using a General Electric Type 213 light meter or equivalent, the light intensity shall be checked at the work floor with the access door closed. The light meter shall be placed on the

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center of the work floor and moved to each corner in turn. The minimum reading shall be 150 foot candles in each corner and 200 foot candles at the center.

*4.5.7 Individual acceptance test. Each production item blast machine shall be subjected to a one-hour performance test using blasting media of size 13 glass beads for 30 minutes and 26 grit aluminum oxide for 30 minutes. During the test period, the blast machine shall perform cleaning tasks typical to the application of this type equipment.

*4.5.8 Noise test (Type I). The noise level at the operator's position shall be measured using a general sound level meter, ANS S1.4-1971 (ANSI Type 2). A noise level in excess of 84 db A slow scale shall cause rejection.

4.6 Inspection of the preservation, packaging, packing and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of Section 5, or the documents specified therein.

5. PACKAGING.

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

*5.1.1 Level A. Each machine shall be preserved and packaged in accordance with MIL-P-116, Method IIA. Accessories shall be individually packaged Method III and consolidated in a domestic fiberboard container conforming to PPP-B-636 and closed with tape conforming to PPP-T-60, Class I. The accessory box shall be secured to the unit within the Method IIA packaged.

*5.1.2 Level C. Each machine shall be preserved and packaged to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the first receiving activity. This level may conform to the supplier's commercial practice when it meets the requirements of this level.

5.2 Packing shall be level A, B or C as specified (see 6.2).

*5.2.1 Level A and B. Each machine preserved and packaged as specified in 5.1.1 shall be packed in a container conforming to MIL-C-52950, Type V, Style B, covered. Interior cushioning, anchoring, blocking and bracing shall be in accordance with MIL-STD-1186. Containers shall be of uniform size and shape and of minimum tare and cube.

*5.2.2 Level C. Each machine preserved and packaged as specified in 5.1.2 shall be packed in a manner which affords adequate protection against damage during shipment from the supply source to the first receiving activity. Containers shall conform to carrier rules and regulations as applicable to the mode of transportation.

5.3 Marking. Unless otherwise specified in the contract or order, interior packages and exterior containers shall be marked in accordance with MIL-STD-129.

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6. NOTES.

*6.1 Intended use. The machines covered by this document are intended for use in cleaning, surface conditioning, and peening of Aerospace Ground Equipment (AGE) and aircraft components.

*6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this document.

b. Type and size required (see 1.2)

c. Level of preservation and packaging, and level of packing required (see 5.1 and 5.2).

6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER AND TITLE
MIL-B-83756C BLAST CLEANING MACHINES

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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